Product Requirement Document (PRD): Industrial Defect Inspection

HIGH-SPEED MANUFACTURING LINE SURFACE QUALITY INSPECTION SPECIFICATION

1. Project Executive Overview

Specification for deploying high-speed computer vision defect classifiers in automotive manufacturing lines across India.

2. Technical & Hardware Constraints Matrix

Parameter	Specification Value	Architectural Note
Model Architecture	ResNet-50 Industrial Defect Classifier	Deep residual convolutional neural network
Model Parameter Count	25.5 Million parameters (0.026B)	Optimized for feature extraction on microscopic surface flaws
Weight Precision	FP16 / Half Precision	Retains micro-crack detection sensitivity without precision clipping
Target Output Time / Latency	25 ms per part (under 30 ms)	Synchronized with industrial assembly conveyor belt speed
Target Throughput	60 FPS (Inspection Frames/Sec)	High-speed line inspection with industrial GigE vision cameras
Target Budget	INR 75,000 (Rs. 75,000)	Factory floor edge box deployment ceiling
Thermal Power Envelope	75 Watts TDP	Passive or DIN-rail fanless enclosure for dusty environments
Make in India Silicon	Medium / Preferred	Evaluating domestic industrial boards and certified rugged accelerators

3. Deployment & Operational Requirements

Deployed across manufacturing shop floors in Pune and Chennai, this automated optical inspection (AOI) system classifies manufactured mechanical parts for micro-fractures and surface defects at 60 FPS. Each image requires decision latency under 25 ms. The edge compute unit operates inside an IP65 sealed enclosure with a thermal ceiling of 75 Watts TDP and a target procurement budget of INR 75,000 (Rs. 75,000).